

Exact value triangles



1 Find the exact value of the following:

a $\sin 30^\circ$

b $\sin 60^\circ$

c $\sin 45^\circ$

d $\cos 45^\circ$

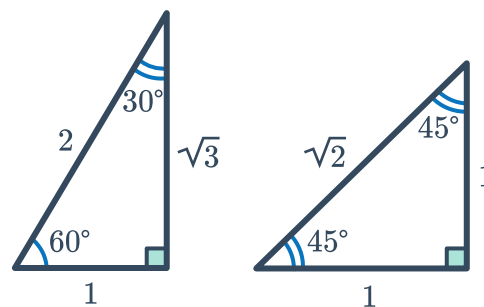
e $\cos 60^\circ$

f $\cos 30^\circ$

g $\tan 60^\circ$

h $\tan 45^\circ$

i $\tan 30^\circ$



2 Evaluate the following expressions, leaving your answer in exact, rationalised form:

a $\frac{\sin 60^\circ}{\cos 30^\circ}$

b $\sin 45^\circ \cos 30^\circ$

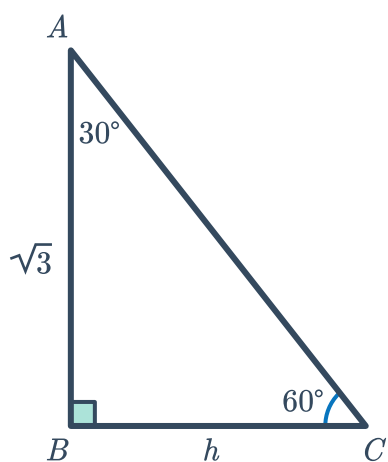
c $\sin 45^\circ + \cos 30^\circ$

d $\sin 45^\circ \cos 60^\circ + \tan 45^\circ$

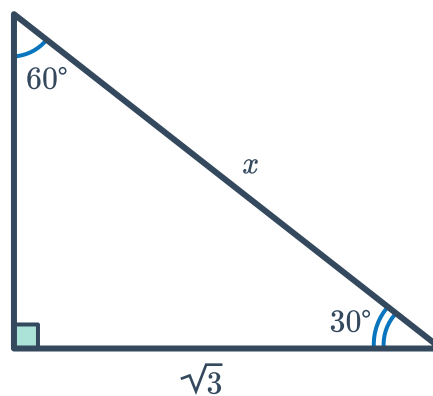
Unknown sides

3 Find the exact value of the pronumeral in the following triangles:

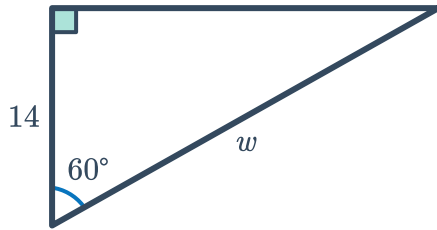
a



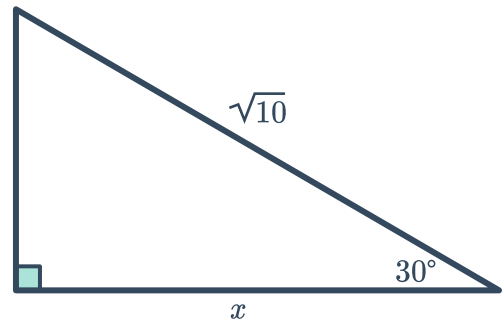
b



c

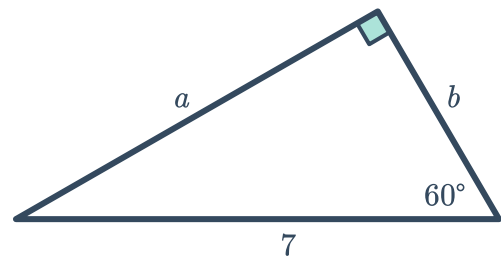


d



4 Consider the given triangle:

- a Find the exact length of side a .
- b Find the exact length of side b .

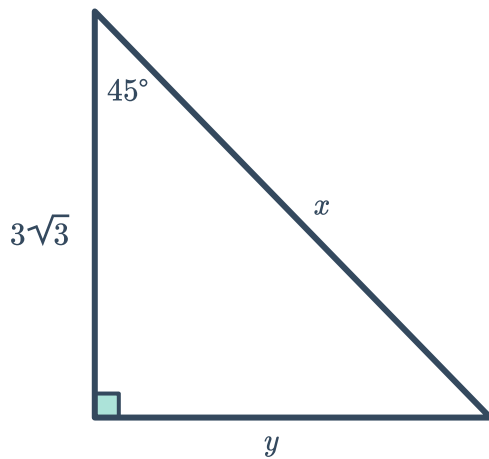


5 For each of the following triangles:

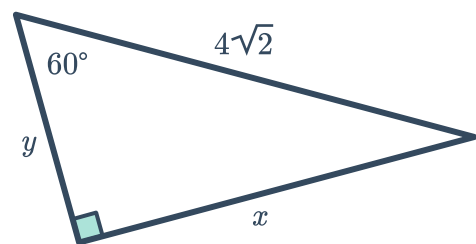
i Find the exact value of x .

ii Find the exact value of y .

a

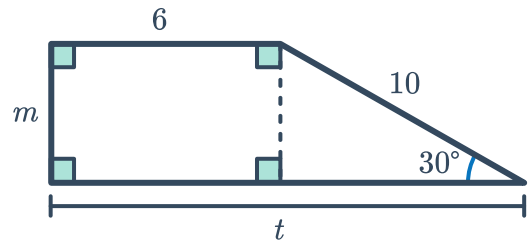


b



6 Consider the unknown lengths in the given figure:

- a Find m .
- b Find t .



Unknown angles

7 θ is an angle in a right-angled triangle. Find the value of θ in the following equations:

a $\cos \theta = \frac{1}{2}$

b $\sin \theta = \frac{\sqrt{3}}{2}$

c $\sin \theta = \frac{1}{\sqrt{2}}$

d $\tan \theta = \frac{1}{\sqrt{3}}$

8 Find the value of θ if $\cos \theta = \frac{1}{\sqrt{2}}$ and $\sin \theta = \frac{1}{\sqrt{2}}$.

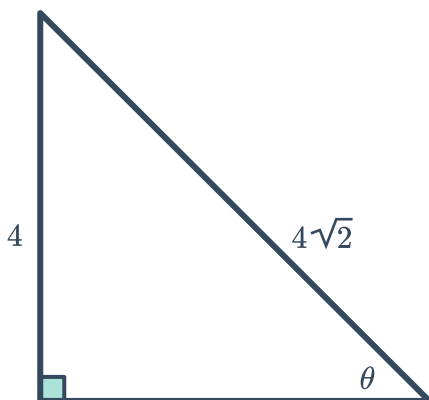
9 Given that $\cos \theta = \frac{\sqrt{3}}{2}$ and $\sin \theta = \frac{1}{2}$:

a Find the value of θ .

b Find the value of $\tan \theta$.

10 Find the unknown θ in the following triangles:

a



b

